Lec 01: Introduction and Proofs

⇒ is not divisible by any other number besides itself and 1

$\forall n \in \mathbb{N}$, n^2+n+41 is a prime number.

⇒ for All

⇒ $\{0, 1, 2, 3 \cdots\}$

Ex) 1 3 5 7 ~~9~~ ~Prime number.

Prime number

$\boxed{\text{Predicate}}$: Proposition whose truth depends on the value of a variable

n	n^2+n+41	Prime
0	41	✓
1	43	✓
2	47	✓
3	53	✓
⋮	⋮	
20	461	✓
⋮	⋮	
39	1601	✓

Ex) $a^2+b^2+c^2=d^2$

has no positive integer solutions

⇒ $a = 95800$
$b = 217519$
$c = 414560$
$d = 422481$

$\Rightarrow$ implies

$\forall n \in \underbrace{\mathbb{Z}}_{\text{int}},\ n \geq 2 \ \boxed{\Rightarrow}\ n^2 \geq 4$

· Def : implies

: An implication $p \Rightarrow q$ is true if p is false OR q is true

Ex) $\forall n \in \mathbb{Z},\ n \geq 2 \underset{\text{iff}}{\Leftrightarrow} n^2 \geq 4$ ... false

$\Leftrightarrow$: $\Rightarrow$, $\Leftarrow$

· truth table.

p	q	$p \Rightarrow q$	$q \Rightarrow p$	$p \Leftrightarrow q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

axioms

Def: An axiom is a proposition that is "assumed" to be true

Ex) If $a=b$ and $b=c$, Then $a=c$

axioms should be

- Consistent
- Complete

Def: A set of axioms is Consistent if no proposition Can be proved to be both true and false.

Def: A set of axioms is Complete if it can be used to prove every proposition is True or False